

WEEK 1 DAY 3

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GROUP:S2

Day 3: Research Framing & Mentor Review

Completed Work So Far

- **Day 1 – Problem Understanding & System Definition**
- Read and rewrote problem statement in own words
- Identified inputs & targets
- Created end-to-end software pipeline
- Defined success criteria
- Prepared 6 questions for mentor
- **Day 2 – Literature Review**
- Reviewed 8 methods/prior systems
- Created comparison table
- Shortlisted candidate methods
- Identified best approaches for our problem

Refined Research Question

models—specifically tree-based algorithms like Random Forest and XGBoost—predict Battery State of Charge (SoC) and Remaining Charging Time using historical charging parameters (voltage, current, power, and temperature), and what are the key features and limitations affecting model performance across different charging phases?"

- **Measurable Outputs:**

- SoC Prediction (MAE < 5%, $R^2 > 0.80$)
- Time Prediction (MAE < 10 min, $R^2 > 0.70$)
- Feature Importance Analysis
- Error Analysis & Failure Cases
- Model Comparison Table

Key Findings from Literature

- Tree-based models (RF, XGBoost) outperform simpler models (Linear).
- Feature engineering is often more important than model selection.
- The “taper effect” (current decreasing near 80% SoC) is the main non-linear challenge.
- Real-world data is noisy – robust preprocessing is essential.

Shortlisted Methods

Baseline Methods:

- Linear Regression

Main Methods:

- Random Forest
- XGBoost

Optional Methods:

- ANN

Key Hypotheses

- XGBoost with robust feature engineering will provide the most accurate predictions.
- Model performance will be best in middle SoC range (30–80%) and worst at extremes (<10% and >90%).
- Temperature will be a significant predictor, especially at high SoC.
- Feature engineering will improve performance by more than 15% compared to raw features.

Proposed Methodology Summary

- **Data Preparation Phase:**

- Load & Inspect Data
- Data Quality Assessment (missing, outliers)
- Data Cleaning & Preprocessing
- Exploratory Data Analysis (EDA)
- Feature Engineering
- Data Splitting (80/20) & Scaling

- **Modeling Phase:**
 - Main Models (Random Forest, XGBoost)
 - Model Tuning (Hyperparameter Optimization)
- **Evaluation Phase:**
 - Evaluation (MAE, RMSE, R^2)
 - Analysis (Feature Importance, Error Analysis)

Evaluation Metrics & Success Criteria

- **Metrics:**

- MAE (Mean Absolute Error)
- RMSE (Root Mean Squared Error)
- R^2 (R-Squared)

- **Success Criteria:**

- SoC: $MAE < 5\%$, $RMSE < 8\%$, $R^2 > 0.80$
- Time: $MAE < 10 \text{ min}$, $RMSE < 15 \text{ min}$, $R^2 > 0.70$

THANK YOU